SCFC · Masters Program v3 — Stanford-Grade Upgrade Spec

> **Document type**: Drop-in upgrade specification for an autonomous coding agent. **Pre-requisite**: `scfc-masters-spec.md` (v2) and `scfc-masters.html` (reference) must be in hand. This document does not restate v2; it specifies what changes. **Output target**: A modified single HTML file (or split per v2 §11.2) that implements the v3 changes additively. Existing v2 progress should be migrated, not destroyed. **Scope**: Stanford MS&E + GSB OIT synthesis, mid-career operator audience, 8–12 week target, mix of worked examples and full derivations on capstone days, expansion from 12 to 18 domains.

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0 · How this document relates to v2

v2 shipped with 60 days, 12 domains, 12 cases, 50 glossary terms, 22 reading entries. v3 keeps that engine and extends it. The rules are:

- **Do not break v2.** The existing `localStorage` key `scfc-masters-v2` migrates with a one-time transformation to `scfc-masters-v3`. Existing user progress (streak, completion history, day position) is preserved.
- **The engine is unchanged in shape.** The state machine (`§3` of v2 spec), persistence schema (`§8`), and component CSS (`§7`) all remain. v3 adds *fields* to existing schemas and *content* to existing arrays — no architectural changes.
- **The visual design is unchanged.** No new colors, no new fonts, no new layouts. The Machine aesthetic is locked.
- **What changes**:
  1. Domain count: 12 → 18 (§2 below)
  2. Curriculum count: 60 → ~80 days, with 8 marked as capstones (§3)
  3. Glossary: +30 terms (§4)
  4. Reading list: +18 entries (§5)
  5. Schema additions: `cap`, `dur`, `deriv` fields on Day; `tag` field on Reading entry (§6)

6. UI additions: capstone-day badge, derivation panel, “FOUNDATION 60” vs “MASTERY +20” mode toggle (§7)
7. Migration logic from v2 state (§8)

If v2 says one thing and this document says another, this document wins for v3.

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1 · Why these changes

The current v2 curriculum maps approximately to MS&E 260 (Introduction to Operations Management) and the operations portion of OIT 262 — solid undergrad / first-year-MBA territory. To reach the breadth and depth of a Stanford MS in supply chain (i.e., MS&E coursework plus elective OIT coursework), the curriculum needs:

More domains. Stanford MS&E and GSB OIT collectively cover six areas v2 doesn’t touch with discipline:

- **Operations Research / Network Optimization** — the spine of MS&E 211, 212, 260, OIT 245/247/248. Linear programming intuition, facility location MILP, vehicle routing. v2 mentions LP in one lecture day; the canon deserves a domain.
- **Sustainability & Climate Operations** — Erica Plambeck’s home territory at Stanford. Scope 3 carbon, supplier emissions, operational hedging against climate risk, the Walmart sustainability index. v2 has one lecture day on Scope 3; the field has a decade of canonical work behind it.
- **Geopolitics & Industrial Policy** — CHIPS Act, IRA, friend-shoring, decoupling. None of this existed in v1 textbooks. It’s what supply chain operators actually face in 2026.
- **Healthcare & Pharma Supply Chain** — MS&E 263 (Healthcare Operations) is a full Stanford course. Cold chain, DSCSA serialization, FDA compliance, vaccine distribution. None in v2.
- **Semiconductor Supply Chain** — TSMC concentration risk, allocation under shortage, lead times of 26+ weeks, the lithography monopoly. The most-discussed supply chain on earth right now, and v2 doesn’t mention it.
- **Behavioral Operations** — Sterman’s beer game, Croson on bullwhip behavior, Gino-Pisano on heuristics, Bendoly’s bodies of knowledge. Stanford OIT and MIT Sloan both teach this; v2 mentions bullwhip’s behavioral roots but doesn’t develop the field.

More math. v2 has three quantitative panels (EOQ, newsvendor, EVPI). A real Stanford MS student derives these results, doesn’t just compute them. v3 adds a derivation panel for ~8 capstone days where the math is the lesson, not the answer. These days are 15–20 minutes

(vs the 5-minute default), explicitly marked, and optional in the sense that progressing past them requires only completing them, not deriving from memory.

More breadth in the canon. v2 cites 22 works. v3 adds 18 more, weighted toward Stanford faculty and OIT-area canonical research: Hau Lee on Triple-A, Plambeck on sustainability, Bimpikis on COVID supply, Sterman on systems dynamics, Pisano on capabilities, Brandenburger-Stuart on value capture.

Honest about what 8–12 weeks delivers. A Stanford MS is two years of full-time study. SCFC delivers, at the new bar, a working principal's *fluency* across that body of knowledge — not the proofs, the published papers, or the research career. The product positions as "Foundation 60" (the existing v2 60 days, slightly reorganized) plus "Mastery +20" (an optional extension covering the new domains and capstones). A user who completes both has 80 days of structured engagement against the canon.

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2 · Domain restructure

v2 had 12 domains. v3 has 18. The reorganization preserves all v2 content under new domain numbers, then adds 6 domains with new days.

2.1 v3 domain list

```
const DOMAINS = [
  { n: 1,  name: 'FOUNDATIONS' },     // unchanged from v2
  { n: 2,  name: 'DEMAND'      },     // unchanged
  { n: 3,  name: 'INVENTORY'   },     // unchanged
  { n: 4,  name: 'SOURCING'    },     // unchanged
  { n: 5,  name: 'CONTRACTS'   },     // unchanged
  { n: 6,  name: 'SUPPLIERS'   },     // unchanged
  { n: 7,  name: 'LOGISTICS'   },     // unchanged
  { n: 8,  name: 'COMMODITY'   },     // unchanged
  { n: 9,  name: 'RISK'        },     // unchanged
  { n: 10, name: 'ANALYTICS'   },     // narrowed; new OR domain takes some conte
  { n: 11, name: 'STRATEGY'    },     // unchanged
  { n: 12, name: 'APPLIED'     },     // unchanged
  // NEW DOMAINS BELOW
  { n: 13, name: 'NETWORKS'    },     // OR / network optimization / facility loc
  { n: 14, name: 'SUSTAIN'     },     // sustainability / climate / Scope 3 / cap
  { n: 15, name: 'GEOPOLITICS' },     // CHIPS Act, IRA, friend-shoring, decoupli
  { n: 16, name: 'HEALTHCARE'  },     // pharma cold chain, DSCSA, vaccine distri
  { n: 17, name: 'SEMICOND'    },     // semiconductor SC, allocation, lithograph
  { n: 18, name: 'BEHAVIOR'    },     // beer game, Sterman, Gino-Pisano, anchori
];
```

The VU meter row count goes from 12 to 18. The CSS for `.vu-mini` already supports arbitrary row counts; no style change required, but the meters block needs to scroll if the panel becomes too tall on mobile. Add `max-height: 280px; overflow-y: auto;` to `.meters-mini` on mobile breakpoints (≤720px).

2.2 Two-tier program structure

The 80 days are presented as two tiers. The toggle UI is a small badge on the nameplate (`§7.4`).

- **Foundation 60** (Days 1–60): the v2 curriculum, slightly reorganized so days 56–60 (which were "Applied" in v2) become the lead-ins to the Mastery extension. The existing reference build is the Foundation tier verbatim.
- **Mastery +20** (Days 61–80): the new content. Six new domains × ~3 days each, plus 1–2 capstone days per new domain. A user must complete Foundation before Mastery is unlockable (mode toggle is locked until day 60 is complete).

The user can replay Foundation days from the curriculum grid at any time. Mastery days appear in the grid only after Foundation completion.

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3 · Curriculum: the new 20 days

> Each day below follows the schema documented in v2 §6.2, plus the new fields described in §6 below.

Day 61 — Triple-A Supply Chain *(case)*

- **dom**: 1 (FOUNDATIONS — extends v2)
- **f**: Triple-A Framework · Agility, Adaptability, Alignment
- **o**: Hau Lee (HBR 2004)
- **cite**: Lee, H. L. "The Triple-A Supply Chain." Harvard Business Review, October 2004.
- **sum**: Speed and cost efficiency are not enough. The best supply chains are simultaneously *agile* (respond fast to short-term changes), *adaptable* (evolve as markets evolve), and *aligned* (incentives match across partners). The framework explains why high-performing chains beat efficient-but-fragile ones.
- **dur**: 5
- **case** "**The Three-A Stress Test**":

- **pre**: A category leader presents your chicken supply network as “world-class.” You are asked to grade it on Lee’s three As before you sign off.
- **setup lines**:
  - `// network audit:`
  - `// AGILITY · response to 2-wk demand shock`
  - `// ADAPT · evolution over 5-yr horizon`
  - `// ALIGN · incentive structure across tiers`
  - blank
  - `// observed:`
  - `// 8-wk lead time (poor agility)`
  - `// no sustainability metric (poor adapt)`
  - `// tier-1 paid on volume only (poor align)`
  - blank
  - `// "world-class" — really?` [amber]
- **ctx**: AGILITY=8 wk · ADAPT=none · ALIGN=volume-only · TIER=Strategic
- **D1** “WHICH A IS WEAKEST.”:
  - **good +2** “ALIGNMENT — incentives drive everything else” → “lee’s most subtle insight. agility and adaptability are constrained by alignment. fix incentives first, the others follow. the chicken steering team has been treating agility as the bottleneck for two years.”
  - **warn 0** “AGILITY — 8-week lead time is the binding constraint” → “plausible at first glance, but the 8-week lead time is itself a consequence of misaligned incentives — tier-1 has no reason to invest in flexibility because they’re paid on volume. you’re treating a symptom.”
  - **warn -1** “ADAPTABILITY — no sustainability tracking is a 5-year problem” → “true but not the binding constraint right now. adapt without align and you’ll evolve in the wrong direction.”
- **D2** “WHAT FIX.”:
  - **good +2** “REWORK INCENTIVE STRUCTURE / GAINSHARING ON FLEX” → “lee’s

prescribed move. tier-1 gets paid for flex performance (lead time variance, fill rate) not just volume. you've changed the equilibrium."

- **bad −2** "INVEST IN AGILITY DIRECTLY / NEW CAPACITY" → "$15M capex on agility a misaligned supplier won't sustain. they'll absorb it as margin and tell you they're agile next quarter while behaving the same."
- **warn 0** "ANNUAL SCORECARD ON ALL THREE AS" → "useful but doesn't change behavior on its own. lee found that scorecards without compensation linkages produce zero behavioral change."

- **opt**: [0, 0]
- **fr**: HAU LEE'S TRIPLE-A (2004) IS THE STANFORD CANON ON OPERATIONAL EXCELLENCE BEYOND COST. THE INSIGHT IS THAT THE THREE AS ARE LEXICOGRAPHICALLY ORDERED WHEN THEY CONFLICT — ALIGNMENT GATES ADAPTABILITY GATES AGILITY. A SUPPLY CHAIN THAT IS FAST AND FLEXIBLE BUT MISALIGNED IS A SUPPLY CHAIN THAT WILL PRODUCE THE WRONG OUTCOME QUICKLY. THE PRINCIPAL CHECKS ALIGNMENT BEFORE INVESTING IN SPEED.

Day 62 — Newsvendor Derivation *(capstone, lecture+derivation)*

- **dom**: 3 (INVENTORY — extends v2)
- **f**: Newsvendor · First-Order Optimality
- **o**: Edgeworth (1888); Arrow, Harris & Marschak (1951)
- **cite**: Arrow, K. J., Harris, T. & Marschak, J. "Optimal Inventory Policy." *Econometrica* 19(3), 1951.
- **sum**: The critical fractile result Q* such that F(Q*) = Cu/(Cu+Co) is not magic. It falls out of differentiating the expected cost function and setting it to zero. Doing the derivation once changes how you think about every stocking problem.
- **dur**: 18
- **cap**: true
- **deriv**: full derivation (see schema §6.3 below for format)
- The v2 day 12 case (newsvendor promo buy) remains. This new day 62 is its companion derivation, optional but recommended.
- **content** (lecture-style, with derivation panel):
  - **brief**: We've used the newsvendor formula. Now we prove it. This is one of the

cleanest applications of first-order optimality conditions in operations research. If you can derive it, you can derive its variants — multi-item, capacity-constrained, price-dependent — without help.

- **derivation steps** (rendered as the new derivation panel):

  1. **Setup**: Demand D is a continuous random variable with CDF F(d) and density f(d). Order quantity Q is the decision. Per-unit underage cost is Cu (margin lost when D > Q). Per-unit overage cost is Co (scrap cost when D < Q).
  2. **Expected cost**: For a given Q,

     ```
     E[C(Q)] = Co · ∫₀^Q (Q − d) f(d) dd  +  Cu · ∫_Q^∞ (d − Q) f(d) dd
     ```

  3. **Differentiate w.r.t. Q** (Leibniz rule):

     ```
     dE[C]/dQ = Co · F(Q)  −  Cu · (1 − F(Q))
              = (Co + Cu) · F(Q)  −  Cu
     ```

  4. **First-order condition**: Set dE[C]/dQ = 0:

     ```
     (Co + Cu) · F(Q*) = Cu
     F(Q*) = Cu / (Co + Cu)          ← the critical fractile
     ```

  5. **Second-order check**: $d^2E[C]/dQ^2 = (Co + Cu) \cdot f(Q) \geq 0$, so Q* is a minimum.
  6. **Interpretation**: F(Q*) is the *probability of not stocking out* at Q*. It equals the ratio of the cost of one too few to the cost of one too few plus one too many. When Cu >> Co, F(Q*) → 1 (stock high). When Co >> Cu, F(Q*) → 0 (stock low).

- **principal framing** (replaces fr): THE DERIVATION IS WORTH MORE THAN THE FORMULA. ONCE YOU'VE DIFFERENTIATED THE NEWSVENDOR EXPECTED-COST FUNCTION, EVERY VARIANT — DEMAND CORRELATED ACROSS PRODUCTS, PRICE-DEPENDENT DEMAND, BACKLOGGABLE SHORTAGE, MULTI-PERIOD INVENTORY — IS A FAMILIAR EXTENSION RATHER THAN A NEW MYSTERY. THIS IS THE DIFFERENCE BETWEEN A USER OF OPERATIONS RESEARCH AND A PRACTITIONER OF IT. ARROW, HARRIS & MARSCHAK (1951) FORMALIZED THE PROOF; THE PRINCIPAL DOES IT ONCE EVERY FEW YEARS TO STAY HONEST.

Day 63 — Network Optimization Intuition *(case)*

- **dom**: 13 (NETWORKS — new)
- **f**: p-Median Facility Location · MILP

- **o**: Hakimi (1964); ReVelle & Swain (1970); Daskin (2013)
- **cite**: Daskin, M. S. *Network and Discrete Location: Models, Algorithms, and Applications.* 2nd ed., Wiley, 2013.
- **sum**: The p-median problem chooses p facility locations from a set of candidates to minimize total demand-weighted distance to customers. It's the foundational supply chain network design model and a binding constraint on every DC discussion.
- **dur**: 8
- **case** "**The Three-DC Question**":
  - **pre**: VP of distribution wants to know whether 2 DCs (current) or 3 DCs (proposal) is right. CFO wants the math. You have a candidate set of 8 cities and demand by region.
  - **setup lines**:
    - `// model: p-median MILP`
    - `// candidates: 8 cities`
    - `// demand: 12 regions, weighted by volume`
    - `// atlanta 22%`
    - `// chicago 14%`
    - `// dallas 11%`
    - `// ... (9 more regions)`
    - blank
    - `// objective: min Σ (demand_i × distance_i,j) for chosen j`
    - `// constraint: exactly p facilities open`
    - blank
    - `// solver returns:`
    - `// p=2: $48M/yr · avg dist 612 mi`
    - `// p=3: $44M/yr · avg dist 421 mi`
    - `// p=4: $43M/yr · avg dist 358 mi`
  - **ctx**: P=2,3,4 · CURRENT=$48M · MIN=$43M · LOAD=600M lb/yr

- **quant** (worked):
  - **f**: `min Σ_i Σ_j d_i,j · h_i · y_i,j s.t. Σ_j x_j = p, y_i,j ≤ x_j`
  - **vars**:
    - [d_i,j, distance from demand i to facility j, miles]
    - [h_i, demand at i, % of total]
    - [x_j, facility j open (0/1), binary]
    - [y_i,j, fraction of i served from j, [0,1]]
  - **r**: `Optimal at p=3: Atlanta + Dallas + Chicago. Marginal value of 4th DC: $1M/yr — below NPV threshold.`
- **D1** "WHAT P.":
  - **good +2** "P=3 · MARGINAL ANALYSIS" → "correct. the p=4 solution saves only $1M more than p=3 against ~$8M new capex. net NPV negative. classical decreasing-marginal-returns network design."
  - **warn 0** "P=4 · LOWEST AVERAGE DISTANCE" → "you optimized for the wrong objective. average distance is a proxy; the cost function already weights it. the marginal analysis says $1M/yr ≠ worth $8M capex."
  - **bad –2** "P=2 · PRESERVE THE CURRENT NETWORK" → "you saved capex and accepted $4M/yr in inferior outcomes that compound. five years from now this is a $20M decision against you."
- **D2** "WHICH 3 CITIES.":
  - **good +2** "ATLANTA + DALLAS + CHICAGO" → "the solver result. each captures a high-density region; they're far enough apart to minimize overlap. this is the p-median optimum."
  - **warn 0** "ATLANTA + DALLAS + LOS ANGELES" → "LA is an instinct play. solver shows chicago dominates LA on weighted distance because midwest demand density beats west-coast density at this volume."
- **opt**: [0, 0]
- **fr**: NETWORK DESIGN IS DECADAL. ONCE YOU SET P AND PICK THE LOCATIONS, YOU LIVE WITH THEM FOR 10–20 YEARS. THE P-MEDIAN MODEL FORCES THE QUESTION INTO TRACTABLE TERMS — BUT THE SOLVER'S ANSWER IS ONLY AS GOOD AS THE DEMAND FORECAST AND COST PARAMETERS YOU FED IT. DASKIN (2013) DEVELOPS THE FAMILY: P-MEDIAN, P-CENTER, COVERING. THE PRINCIPAL

UNDERSTANDS WHICH FORMULATION FITS WHICH PROBLEM AND DOES MARGINAL ANALYSIS BEFORE COMMITTING CAPEX.

Day 64 — Vehicle Routing Problem *(lecture)*

- **dom**: 13 (NETWORKS)
- **f**: VRP · NP-Hard Combinatorial Optimization
- **o**: Dantzig & Ramser (1959); Toth & Vigo (2014)
- **cite**: Toth, P. & Vigo, D. *Vehicle Routing: Problems, Methods, and Applications*. SIAM, 2014.
- **sum**: The Vehicle Routing Problem assigns a fleet of vehicles to customer locations such that total distance/time is minimized subject to capacity, time-window, and driver-shift constraints. NP-hard in general — practitioners use heuristics (savings algorithm, tabu search, large neighborhood search). Solving VRP well at scale is worth single-digit millions per year for any large logistics network.
- **dur**: 5

Day 65 — Queueing Intuition · Little's Law *(lecture)*

- **dom**: 10 (ANALYTICS)
- **f**: Little's Law · $L = \lambda \times W$
- **o**: Little (1961); Erlang (1909) for the discipline
- **cite**: Little, J. D. C. "A Proof for the Queuing Formula: $L = \lambda W$." *Operations Research* 9(3), 1961.
- **sum**: For any stable system, average number in system = arrival rate × average time in system. Holds for manufacturing WIP, customer service queues, hospital throughput, even API request handling. The most-used result in operations research.
- **dur**: 5

Day 66 — Queueing Capstone *(capstone)*

- **dom**: 10 (ANALYTICS)
- **f**: M/M/1, M/M/c, Erlang C
- **o**: Erlang (1909); Allen-Cunneen for approximations
- **cite**: Hopp, W. J. & Spearman, M. L. *Factory Physics*. 3rd ed., 2008. (Ch. 8)

- **sum**: How many docks does a DC need? How many service desks at peak? Erlang's queueing models give defensible answers. The math is non-trivial but the intuition is clean: utilization × variability = wait. As ρ approaches 1, wait blows up.
- **dur**: 18
- **cap**: true
- **deriv**: full
- **content** (lecture+derivation):
  - **brief**: A DC has a single inbound dock receiving trucks. Arrivals are roughly Poisson, service is roughly exponential. How does wait time scale with utilization?
  - **derivation steps**:
    1. **Setup**: M/M/1 queue. Arrivals λ (per hour), service rate μ (per hour). Utilization ρ = λ/μ. Stable if ρ < 1.
    2. **State probabilities** (steady-state, derive from balance equations):

       ```
       π_n = ρ^n · (1 − ρ),   for n = 0, 1, 2, ...
       ```

    3. **Average number in system**:

       ```
       L = Σ n · π_n = ρ / (1 − ρ)
       ```

       Derivation uses Σ n·ρ^n = ρ/(1−ρ)² for |ρ| < 1.
    4. **Apply Little's Law** (L = λW):

       ```
       W = L / λ = 1 / (μ(1 − ρ))
       ```

    5. **Wait time in queue alone (excluding service)**:

       ```
       Wq = ρ / (μ(1 − ρ)) = (1/μ) · ρ/(1−ρ)
       ```

    6. **The pole at ρ = 1**: wait time goes to infinity as utilization approaches 1. This is the *fundamental tradeoff* of operations: you cannot run at 100% utilization and have low wait. Variability requires slack.
    7. **Numerical example**: ρ = 0.7 → average wait ≈ 2.33 service times. ρ = 0.9 → 9.0. ρ = 0.95 → 19.0. The cost of pushing utilization is non-linear.

- **fr**: ERLANG (1909) AND LITTLE (1961) GAVE OPERATIONS THE QUEUEING CANON. THE PRACTICAL INSIGHT THAT EVERY MANAGER MUST INTERNALIZE: WAIT TIME IS NOT LINEAR IN UTILIZATION — IT IS HYPERBOLIC. RUNNING A DC AT 95% UTILIZATION DOES NOT COST 5% MORE WAIT THAN 90% — IT COSTS 2× MORE. ANY SCHEDULER WHO PUSHES UTILIZATION TO THE CEILING WITHOUT QUANTIFYING THE QUEUEING COST IS DOING ARITHMETIC AGAINST PHYSICS. THE PRINCIPAL TARGETS UTILIZATION IN THE 70–85% BAND FOR VARIABLE SYSTEMS, ACCEPTS HIGHER ONLY WHERE VARIABILITY IS LOW.

Day 67 — Sustainability & Operations *(case)*

- **dom**: 14 (SUSTAIN — new)
- **f**: Operational Sustainability · Triple Bottom Line in OM
- **o**: Plambeck (Stanford); Lee, Plambeck & Yatsko on Walmart
- **cite**: Plambeck, E. L. "Reducing greenhouse gas emissions through operations and supply chain management." *Energy Economics* 34, 2012.
- **sum**: The 80%+ of corporate emissions that live in Scope 3 (the supply chain) cannot be reduced by buying renewable energy at headquarters. They require operational changes — supplier selection, transportation mode, packaging weight, demand smoothing. Stanford's Plambeck has documented the playbook over 20 years.
- **dur**: 8
- **case** "**The Walmart Question**":
  - **pre**: An executive memo asks whether Chick-fil-A should adopt a "supplier sustainability scorecard" similar to Walmart's. Operations is asked for a recommendation in 2 weeks.
  - **setup lines**:
    - `// scope 3 emissions:`
    - `// purchased goods 62% (cat. 1)`
    - `// transport 18% (cat. 4)`
    - `// capital goods 7% (cat. 2)`
    - `// ... (12 other categories)`
    - blank
    - `// ~80% of the company's footprint is upstream.`

- `// internal energy efficiency: ~3% of total.`
- blank
- `// walmart scorecard reduced supplier emissions ~20%`
- `// over 8 years (lee, plambeck, yatsko 2012)`

- **ctx**: SCOPE 3=80% · CAT 1=62% · CAT 4=18% · WALMART REDUCTION=20%/8yr
- **D1** "PRIMARY MOVE.":
  - **good +2** "SUPPLIER SCORECARD W/ EMISSIONS METRIC + COMPENSATION LINKAGE" → "plambeck-style. you measure, you pay for it. the walmart program achieved 20% reduction over 8 years. the alternative (declarations without compensation) achieves nothing."
  - **warn 0** "MEASURE EMISSIONS / NO COMPENSATION LINKAGE" → "you've created a reporting overhead. plambeck's research shows declarations without incentives produce 0–3% reduction. the supplier optimizes against what you pay for."
  - **bad −2** "INVEST IN ON-SITE RENEWABLE ENERGY" → "you addressed scope 1+2 (the 20%) and ignored scope 3 (the 80%). this is the most common corporate sustainability error. you'll publish a report and your real footprint will be unchanged."
- **D2** "TIMELINE EXPECTATION.":
  - **good +2** "5–10 YEARS · GRADUAL · EMBED IN QBR" → "plambeck's data shows 20% reduction takes 8 years even with disciplined execution. shorter timelines are corporate theater."
  - **warn 0** "3 YEARS · QUARTERLY MILESTONES" → "you set up 12 quarterly failures. the supplier base cannot move that fast on capital decisions (boilers, fleet, packaging lines). you'll be back at the table in year 2 explaining why."
  - **bad −2** "ANNUAL REPORTING / NO TIMELINE COMMITMENT" → "you bought a clean PR cycle. when the next ESG report cycle hits, you'll be the company without quantified targets and the regulators are increasingly noticing."
- **opt**: [0, 0]
- **fr**: PLAMBECK'S BODY OF WORK (~2005–PRESENT) ESTABLISHED THAT OPERATIONAL SUSTAINABILITY IS A SUPPLY CHAIN PROBLEM, NOT AN ENERGY PROCUREMENT ONE. THE WALMART SCORECARD CASE IS THE CANONICAL EXAMPLE: A LARGE BUYER USING SUPPLIER COMPENSATION TO MOVE THE TIER-

N FOOTPRINT OF AN ENTIRE INDUSTRY. THE PRINCIPAL UNDERSTANDS THAT THE 80% IN SCOPE 3 IS WHERE THE ACTUAL CARBON LIVES, AND THAT MOVING IT REQUIRES PUTTING THE METRIC INTO THE SUPPLIER SCORECARD WITH MONEY ATTACHED.

Day 68 — Carbon Pricing & Cap-and-Trade *(lecture)*

- **dom**: 14 (SUSTAIN)
- **f**: Carbon Markets · EU ETS · CBAM
- **o**: EU ETS Phase IV (2021); CBAM (2023)
- **cite**: European Commission, "Carbon Border Adjustment Mechanism," Regulation (EU) 2023/956.
- **sum**: Carbon pricing is no longer abstract. The EU charges €80–100/tonne CO_2 via ETS; CBAM (Carbon Border Adjustment) prices imported carbon as well. For US firms exporting to EU or sourcing from EU, this is a real cost on the P&L. The principal models it as a commodity with futures, basis risk, and pass-through dynamics — exactly v2's commodity domain framework.
- **dur**: 5

Day 69 — Sustainability Capstone · Carbon Math *(capstone)*

- **dom**: 14 (SUSTAIN)
- **f**: Activity-Based Scope 3 Calculation
- **o**: WRI/WBCSD GHG Protocol Scope 3 Standard (2011)
- **cite**: GHG Protocol. *Corporate Value Chain (Scope 3) Accounting and Reporting Standard*. WRI/WBCSD, 2011.
- **sum**: Calculating Scope 3 from real procurement data — not estimates, not industry averages. The Standard defines five calculation methods; the principal picks the right one based on data quality and audit defensibility.
- **dur**: 18
- **cap**: true
- **deriv**: full
- **content**:
  - **brief**: You receive 480,000 cases of shell eggs annually from a single supplier. EPA

emission factors and supplier-specific energy data are both available. How do you compute the Scope 3 Category 1 (“Purchased Goods”) emissions?

- **derivation steps**:

  1. **Method hierarchy** (per GHG Protocol Scope 3 §7.3.4):
     - Method 1: Supplier-specific (highest quality)
     - Method 2: Hybrid (supplier-specific + average)
     - Method 3: Average-data (industry emission factor × physical units)
     - Method 4: Spend-based (industry factor × $ spent)
  2. **Method 3 calculation** (most common in practice):

     ```
     Emissions (tCO₂e) = Activity × Emission Factor
                       = 480,000 cases × 30 dozen/case × 12 eggs/dozen
                          × 4.8 g CO₂e/egg (USDA EPA factor for shell eggs)
                       = 480,000 × 360 × 4.8 g
                       = 829,440,000 g
                       = 829 tCO₂e/yr
     ```

  3. **Method 1 calculation** (supplier-specific, from supplier’s actual energy/feed data):

     ```
     Supplier reports: 712 tCO₂e/yr for your portion of their volume
     (lower because their layer barns are 14% more efficient than industry a
     ```

  4. **Audit defensibility**: Method 1 is preferred when data exists but requires verifiable supplier data. Method 3 is acceptable when supplier data is absent. Method 4 (spend-based) is the lowest tier — used when nothing else is available, but increasingly being rejected by audit firms.
  5. **At the carbon price**: At €80/tCO_2e (current EU ETS), the Method 1 number (712 tCO_2e) implies €56,960/yr in carbon-equivalent cost embedded in your egg supply — small per case (~$0.12) but real, and growing.
  6. **The decision**: When a supplier offers Method 1 data, accept it (it’s lower than the average). When they don’t, push for it — Method 1 is what gets you defensibility and gives you leverage on the next contract.

- **fr**: SCOPE 3 CALCULATION IS NOT AN ESG EXERCISE — IT IS A PROCUREMENT EXERCISE. THE GHG PROTOCOL STANDARD GIVES FOUR METHODS, ORDERED BY DATA QUALITY. THE PRINCIPAL ALWAYS PUSHES FOR METHOD 1 OR 2 BECAUSE

(A) IT IS DEFENSIBLE IN AUDIT, (B) IT GIVES THE CORRECT NUMBER FOR DECISION-MAKING, AND (C) IT CREATES A REASON FOR THE SUPPLIER TO INVEST IN EFFICIENCY (BECAUSE NOW THEY'RE MEASURED ON IT). USING METHOD 4 (SPEND-BASED) IS LIKE USING SHIPPED-WEIGHT FOR PRICING — IT'S THE TOOL OF LAST RESORT, NEVER THE FIRST CHOICE.

Day 70 — Geopolitical Supply Chain *(case)*

- **dom**: 15 (GEOPOLITICS — new)
- **f**: Friend-shoring · Decoupling · Chokepoint Analysis
- **o**: Yellen (2022, friend-shoring); Witt et al. (2023, decoupling)
- **cite**: Witt, M. A., Lewin, A. Y., Li, P. P. & Gaur, A. "Decoupling in international business: Evidence, drivers, impact, and implications." *Journal of World Business* 58(2), 2023.
- **sum**: Geographic concentration is a strategy, not just a fact. The CHIPS Act, IRA, and EU Critical Raw Materials Act are explicitly redirecting supply chain capital. Operators in 2026 must decide whether to chase the subsidies, hedge against decoupling, or hold position.
- **dur**: 8
- **case** "**The Single-Country Sourcing Question**":
  - **pre**: 73% of your specialty packaging comes from a single Chinese supplier. The CHIPS Act and IRA don't apply to your category, but Section 301 tariffs do. Your CFO asks whether to relocate.
  - **setup lines**:
    - `// concentration:`
    - `// chinese supplier 73% ($28M/yr)`
    - `// mexican alt 18% ($7M/yr)`
    - `// us alt 9% ($3.5M/yr)`
    - blank
    - `// section 301 tariff: 25% (currently active)`
    - `// chinese landed cost incl. tariff: still 8% lower than mexico`
    - `// mexico capacity: limited, 6mo to expand`

        - blank
        - `// CFO question: hold or move?`
    - **ctx**: SPEND=$38.5M · CHINA=73% · TARIFF=25% · COST GAP=−8%
    - **D1** "STRATEGY.":
        - **warn -1** "HOLD POSITION · ABSORB TARIFF" → "expected-cost works today (china is still 8% cheaper net). but the tariff trajectory is one-way; section 301 has been raised, never lowered, since 2018. you're betting on a reversal that hasn't happened."
        - **good +2** "MOVE TO 50/40/10 · MEXICO PRIMARY · 18-MO TRANSITION" → "friend-shoring play. you accept short-term margin pressure (mexico is 8% more expensive at lower volume) for medium-term resilience. when section 301 escalates again — which the literature predicts — you're already adjusted."
        - **warn 0** "RESHORE TO US · 100% DOMESTIC" → "over-rotation. the cost gap is too large; you'd give up 22% margin without proportional risk reduction. friend-shoring is a 70/30 play, not a 100% one."
    - **D2** "COMMUNICATION.":
        - **good +2** "FRAME AS RISK MANAGEMENT, NOT POLITICS" → "the witt et al. (2023) decoupling research framing. avoid the political narrative; lead with disruption probability × cost. the audit committee approves on those terms."
        - **bad −2** "FRAME AS GEOPOLITICAL ALIGNMENT" → "you've made a procurement decision political. half your stakeholders disagree with the framing. the decision was right; the framing makes it fragile to leadership turnover."
    - **opt**: [1, 0]
    - **fr**: THE 2018–PRESENT ERA OF INDUSTRIAL POLICY (TRUMP TARIFFS → BIDEN CHIPS/IRA → EU CRITICAL RAW MATERIALS) HAS PERMANENTLY CHANGED SUPPLY CHAIN STRATEGY. SINGLE-COUNTRY CONCENTRATION IS NOW A POLICY-DEPENDENT BET. THE PRINCIPAL'S MOVE IS NOT TO PICK SIDES BUT TO FRAME EVERY GEOGRAPHIC CONCENTRATION DECISION AS RISK MANAGEMENT — DISRUPTION PROBABILITY × DISRUPTION COST — RATHER THAN POLITICS. THIS IS THE FRAMEWORK THAT SURVIVES LEADERSHIP TRANSITIONS AND SCALES TO THE NEXT POLICY CYCLE.

Day 71 — CHIPS Act & Industrial Policy *(lecture)*

- **dom**: 15 (GEOPOLITICS)

- **f**: Industrial Policy as Supply Chain Lever
- **o**: CSIS (2023); Lovely (2023)
- **cite**: Lovely, M. E. "Manufacturing Resilience: The US Drive to Reorder Global Supply Chains." Economic Strategy Group, 2023.
- **sum**: $200B+ CHIPS Act + $370B IRA + Bipartisan Infrastructure Law = ~$100B/yr in industrial subsidies for the next 5 years. The largest peacetime industrial policy intervention in US history. Operators in eligible categories (semis, EV batteries, clean energy, advanced packaging) are restructuring strategy around the subsidy stack.
- **dur**: 5

Day 72 — Healthcare Supply Chain *(case)*

- **dom**: 16 (HEALTHCARE — new)
- **f**: Cold Chain Operations · Pharma Distribution
- **o**: Stanford MS&E 263; Scheinker (2024)
- **cite**: Scheinker, D. & Brandeau, M. L. "Healthcare operations management: A bridge to the future." *Annals of Operations Research*, various.
- **sum**: Pharma cold chain is the most demanding logistics environment in commerce. Failure modes are clinical, not commercial — a temperature excursion doesn't reduce margin, it kills patients. The discipline is: redundant temperature monitoring, validated lanes, and audit trails.
- **dur**: 8
- **case** "**The Vaccine Allocation Question**":
  - **pre**: A specialty distributor is considering a contract with a pharma supplier for cold-chain (2–8°C) vaccine distribution. 6,000 units/month, 4 metro markets. The contract requires audit-ready temperature logs at all transitions.
  - **setup lines**:
    - `// pharma cold chain requirements:`
    - `// temperature: 2–8°C continuous`
    - `// transition logs: audit-ready, signed`
    - `// excursion handling: <5 min before alert`
    - `// recall capability: <24h to all consignees`

        - blank
        - `// volume: 6,000 units/mo, 4 metros`
        - `// margin uplift over std cold chain: 12%`
        - `// capex required: $480k (validated containers + monitoring)`
        - `// liability exposure if excursion: $4–9M per event`
    - **ctx**: VOL=6,000 u/mo · MARGIN=+12% · CAPEX=$480k · LIAB=$4–9M
    - **D1** "TAKE THE CONTRACT.":
        - **warn 0** "TAKE IT · UPSIDE WORTH RISK" → "12% margin uplift on $X is real, but you've contracted to a liability ceiling 8–18× the annual margin gain. one excursion in a 5-year contract destroys the entire economics."
        - **good +2** "TAKE IT · WITH FORMAL RISK MGMT INVESTMENT" → "you've sized the liability. the answer is yes, but only with an additional $200k/yr in monitoring redundancy, dedicated cold-chain ops, and FDA audit prep. net economics still positive after that."
        - **bad –2** "DECLINE · TOO MUCH RISK" → "you walked from $X/yr in margin uplift because you didn't size the risk. healthcare logistics is high-margin precisely because the operational bar is high; declining all of it on principle leaves you in the commodity logistics market."
    - **D2** "MOST-IMPORTANT INVESTMENT.":
        - **good +2** "REDUNDANT TEMPERATURE MONITORING / 24/7 ALERT" → "the FDA expects this. it's also where 80% of preventable excursions are caught. minimum cost, maximum risk reduction."
        - **warn 0** "INSURANCE / EXCURSION COVERAGE" → "addresses the financial tail but not the operational one. an FDA Form 483 is not just a financial event — it shuts down your contract regardless of insurance."
        - **bad –2** "DRIVER TRAINING" → "important but lowest-ROI of the choices. driver error accounts for <10% of pharma cold chain incidents per FDA data."
    - **opt**: [1, 0]
    - **fr**: HEALTHCARE SUPPLY CHAIN IS WHERE OPERATIONS RESEARCH MEETS REGULATORY LIABILITY. THE FDA AND DSCSA HAVE TURNED PHARMA DISTRIBUTION INTO A FORMAL VALIDATION DISCIPLINE — SERIALIZATION, EPCIS DATA EXCHANGE, COLD-CHAIN AUDIT TRAILS. STANFORD MS&E 263

(HEALTHCARE OPERATIONS) DEVELOPS THE QUANTITATIVE TOOLS FOR THIS DOMAIN, AND THE CONSEQUENCE OF GETTING IT WRONG IS NOT MARGIN LOSS BUT PATIENT HARM. THE PRINCIPAL TREATS HEALTHCARE SUPPLY CHAIN AS A SEPARATE DISCIPLINE WITH ITS OWN ECONOMICS, NOT AS A HIGH-MARGIN VARIANT OF GENERAL LOGISTICS.

Day 73 — DSCSA & Pharma Compliance *(lecture)*

- **dom**: 16 (HEALTHCARE)
- **f**: DSCSA · Item-Level Serialization
- **o**: FDA (Drug Quality and Security Act, 2013); fully enforced 2024
- **cite**: FDA. *Drug Supply Chain Security Act (DSCSA): Implementation Guide*. 2024.
- **sum**: As of 2024, every prescription drug in the US has unit-level serialization, end-to-end electronic chain-of-custody (EPCIS), and verifiable transactions at every handoff. The largest mandatory traceability rollout in commerce history. Operators outside healthcare should study it; serialization is coming for food, electronics, and apparel.
- **dur**: 5

Day 74 — Semiconductor Supply Chain *(case)*

- **dom**: 17 (SEMICOND — new)
- **f**: Concentration Risk · Lead Time · Allocation
- **o**: Lee (2022, semiconductor crisis); CSIS (2023)
- **cite**: Allen, G. C. *Sourcing Requirements and U.S. Technological Competitiveness*. CSIS, 2024.
- **sum**: TSMC produces ~60% of advanced semiconductors and ~90% of the most-advanced (≤5nm). ASML's EUV lithography monopoly. Lead times of 26+ weeks during shortage. The most-discussed supply chain in the world; understanding it is operator literacy in 2026.
- **dur**: 8
- **case** "**The Allocation Crisis**":
  - **pre**: A semiconductor-dependent product line gets a "30% allocation" notice from your fab partner — meaning you'll receive 30% of forecasted volume for the next 2 quarters. Engineering says they can re-spec to a less-constrained node, but it costs $2M and 9 months.

- **setup lines**:
  - `// allocation event:`
  - `// forecasted: 480k units/qtr`
  - `// allocated: 144k (30%)`
  - `// gap: 336k/qtr`
  - blank
  - `// re-spec option:`
  - `// move from 5nm → 7nm`
  - `// cost: $2M nre + 9 mo lead`
  - `// 7nm has 2× available capacity`
  - blank
  - `// market: lost units = lost revenue + competitive cost`
  - `// $/unit margin: ~$8`
- **ctx**: GAP=336k/qtr · MARGIN=$8 · NRE=$2M · LEAD=9mo
- **D1** "RESPONSE.":
  - **good +2** "RE-SPEC TO 7NM · ACCEPT 9MO COST" → "9 months × 336k × $8 = $24M in lost margin during transition, but you escape allocation risk for the foreseeable future. fab capacity at older nodes is structurally larger and less geopolitically concentrated."
  - **warn 0** "PAY PREMIUM FOR ADDITIONAL 5NM ALLOCATION" → "the gray market for advanced node allocation runs 2–4× spot. you got the units this quarter but signed up for a ratcheted dependency. next allocation cycle, your premium goes up."
  - **bad −2** "ACCEPT THE ALLOCATION · WAIT IT OUT" → "you absorbed $24M in lost margin for 6 months on the bet that allocation eases. semiconductor allocation cycles run 12–18 months. you bet wrong by an order of magnitude."
- **D2** "LONG-TERM POSTURE.":
  - **good +2** "DUAL-NODE DESIGN · 5NM + 7NM CAPABILITY" → "engineering capability for both nodes means you can flex against allocation in real time. the cost is product complexity; the benefit is permanent allocation insurance."

        - **warn 0** “STRATEGIC PARTNERSHIP / CAPACITY RESERVATION” → “TSMC offers reserved-capacity contracts for $20M+ commitments. works for hyperscalers; for $30M annual customers, it’s a misallocation.”
    - **opt**: [0, 0]
    - **fr**: THE SEMICONDUCTOR INDUSTRY HAS THE MOST CONCENTRATED, MOST GEOGRAPHICALLY-EXPOSED SUPPLY CHAIN ON EARTH. TSMC FOR ADVANCED LOGIC; ASML FOR EUV LITHOGRAPHY; SAMSUNG/MICRON/SK HYNIX FOR DRAM; A HANDFUL OF JAPANESE FIRMS FOR PHOTORESIST. THE ALLOCATION DYNAMICS ARE GAME-THEORETIC — WHEN CAPACITY IS CONSTRAINED, THE FAB ALLOCATES TO ITS LARGEST AND MOST STABLE CUSTOMERS FIRST. THE PRINCIPAL UNDERSTANDS THAT BUYING SEMIS IS NOT BUYING WIDGETS; IT IS BUYING A POSITION IN AN ALLOCATION PRIORITY QUEUE, AND THE COST OF MOVING UP THAT QUEUE IS COMMITMENT VOLUME.

Day 75 — Lithography Monopoly · ASML *(lecture)*

- **dom**: 17 (SEMICOND)
- **f**: Single-Vendor Dependency in High-Tech
- **o**: Industry analysis; Mazzucato on industrial policy
- **cite**: Mazzucato, M. *The Entrepreneurial State: Debunking Public vs Private Sector Myths*. Anthem Press, 2013.
- **sum**: ASML is the sole supplier of EUV lithography machines worldwide. Each machine costs ~$200M, requires 40+ trucks to ship, and is the gating technology for ≤7nm semiconductor production. The most concentrated single-vendor dependency in modern industry — and the geopolitical leverage point of US export controls.
- **dur**: 5

Day 76 — Behavioral Operations *(case)*

- **dom**: 18 (BEHAVIOR — new)
- **f**: The Beer Distribution Game · Sterman’s Findings
- **o**: Sterman (1989); Croson & Donohue (2006)
- **cite**: Sterman, J. D. “Modeling Managerial Behavior: Misperceptions of Feedback in a Dynamic Decision Making Experiment.” *Management Science* 35(3), 1989.
- **sum**: Sterman’s beer game shows that bullwhip emerges even with full information and rational players, because humans systematically misperceive feedback delays. The

implication: bullwhip is partly a behavioral problem and information sharing alone doesn't fix it. Compensation, training, and decision-support tools must address human cognitive limits.

- **dur**: 8
- **case** "**The Order-Up-To Override**":
  - **pre**: Your sourcing team has been running an automated order-up-to inventory policy for 3 years. A new VP is uncomfortable letting an algorithm "make the call" and proposes adding human approval at every reorder. ROI of automation is documented at $4M/yr in carrying-cost reduction.
  - **setup lines**:
    - `// 3-year automation outcome:`
    - `// carrying cost −18% (~$4M/yr)`
    - `// stockouts −12%`
    - `// override frequency 4% (legitimate edge cases)`
    - blank
    - `// proposal: human approval at every reorder`
    - `// expected effect (croson & donohue 2006):`
    - `// inventory overshoot +20–35%`
    - `// stockouts unchanged`
    - `// override frequency 100%`
    - blank
    - `// VP wants discomfort relief, not data`
  - **ctx**: AUTOMATION SAVINGS=$4M/yr · PROPOSED OVERRIDE=100% · OVERSHOOT=+25%
  - **D1** "RESPOND TO THE VP.":
    - **good +2** "PRESENT THE BEHAVIORAL DATA / OFFER REVIEW DASHBOARD" → "sterman & croson research shows humans systematically over-order under uncertainty. you've documented $4M of bullwhip avoidance. offer a review-only dashboard — VP gets visibility without the override authority."

        - **warn 0** "ACCEPT 50% AS COMPROMISE" → "you've replaced 96% automated efficiency with 50% — the math says inventory creep returns at half the rate. compromises that ignore the underlying mechanism don't compromise the mechanism."
        - **bad −2** "DEFER TO VP · INSTITUTE 100% APPROVAL" → "you traded $4M in documented savings for the VP's emotional comfort. when next year's QBR shows working capital up 15%, the algorithm gets blamed. classic behavioral-ops failure mode."
    - **D2** "LONG-TERM FRAMING.":
        - **good +2** "POSITION AS DECISION-SUPPORT, NOT REPLACEMENT" → "gino & pisano framing. the algorithm doesn't replace judgment; it removes the cognitive load of routine decisions so judgment can be applied to genuine edge cases. VP gets agency where it matters."
        - **warn 0** "ESCALATE TO HIGHER LEADERSHIP" → "you've made it political. you may win the argument but lose the relationship. behavioral ops literature consistently finds that operations leaders who ground decisions in research outperform those who ground them in authority."
    - **opt**: [0, 0]
    - **fr**: BEHAVIORAL OPERATIONS (GINO & PISANO 2008, CROSON & DONOHUE 2006, STERMAN 1989) ESTABLISHED THAT THE HUMAN DECISION-MAKER IS NEVER OPTIMAL — AND THAT MOST OPERATIONAL PROBLEMS ARE PARTLY BEHAVIORAL. THE IMPLICATION FOR THE PRINCIPAL IS THAT *DESIGNING THE DECISION ENVIRONMENT* MATTERS AS MUCH AS THE FORMAL OPTIMIZATION. AUTOMATION REMOVES COGNITIVE LOAD; SCORECARDS DIRECT ATTENTION; INFORMATION DISPLAYS CHANGE BEHAVIOR. THE PRINCIPAL APPLIES BEHAVIORAL INSIGHTS *TO* THE OPERATING SYSTEM AS A DESIGN INPUT, NOT JUST AS AN AFTER-THE-FACT EXPLANATION OF FAILURE.

Day 77 — Anchoring & Negotiation *(lecture)*

- **dom**: 18 (BEHAVIOR)
- **f**: Anchoring Bias · First-Offer Effects
- **o**: Tversky & Kahneman (1974); Galinsky & Mussweiler (2001) for negotiation
- **cite**: Galinsky, A. D. & Mussweiler, T. "First Offers as Anchors: The Role of Perspective-Taking and Negotiator Focus." *J. Personality and Social Psych.* 81(4), 2001.
- **sum**: First-offer anchoring is among the most-replicated effects in negotiation research.

Whoever opens the negotiation typically captures 5–15% more value than the responder. Procurement professionals systematically *under-use* this — they wait for the supplier's number. The principal opens.

- **dur**: 5

Day 78 — Capstone · Multi-Domain Synthesis *(capstone)*

- **dom**: 12 (APPLIED — extends v2)
- **f**: Cross-Framework Reasoning Under Constraint
- **o**: Curated synthesis · Stanford OIT canon
- **cite**: Lee, Plambeck, Sheffi, Goldratt, Cachon, Tang, Raiffa, Sterman — selected works.
- **sum**: A real procurement decision pulls from 5+ frameworks simultaneously. This is the integration challenge — not knowing each framework, but knowing which ones apply, in what order, and where they conflict.
- **dur**: 20
- **cap**: true
- **case** "**The Friday Decision**":
  - **pre**: Your top supplier (strategic, $142M, 15-year relationship) is in a public sustainability scandal with regulatory exposure. The CEO needs your recommendation by Friday: maintain, terminate, or restructure. The decision pulls from sourcing, contracts, sustainability, geopolitics, and behavioral framing simultaneously.
  - **setup lines**:
    - `// situation:`
    - `// supplier: Tier-1 strategic, $142M/yr`
    - `// scandal: scope 3 violation, EU CBAM exposure`
    - `// media: 3 articles, 1 European, escalating`
    - `// regulatory: ECHA investigation announced`
    - blank
    - `// frameworks in play:`
    - `// KRALJIC strategic placement → partnership`

        - `// TANG supply concentration → diversify?`
        - `// FISHER-URY BATNA → 9mo to restructure`
        - `// PLAMBECK scope 3 → audit defense`
        - `// STERMAN reactive bullwhip risk`
        - `// SHEFFI resilience flex options`
        - blank
        - `// CEO wants the recommendation. friday.`
    - **ctx**: SPEND=$142M · CONC=high · BATNA=9mo · MEDIA=escalating · REG=open
    - **D1** "FRAMING.":
        - **good +2** "REGULATORY-FIRST · STAGED RESTRUCTURE" → "the cardinal principle: the regulator's clock is faster than yours. lead the framing with regulatory cooperation (CBAM disclosure, supplier audit), then negotiate restructure from that position. preserves the strategic relationship while protecting the reg flank."
        - **warn -1** "MAINTAIN POSITION · WEATHER MEDIA" → "you treated the scandal as a media problem. ECHA investigations don't resolve in news cycles. by week 6 you're either named or you've spent 4 weeks unable to negotiate from strength."
        - **bad –2** "PUBLICLY TERMINATE / SIGNAL VALUES" → "you've solved the optics and created a $142M sourcing crisis with no BATNA. tang's expected-cost math: 9-month transition × peak risk ≈ $80M+ in operational disruption."
    - **D2** "RESTRUCTURE TERMS.":
        - **good +2** "OPEN-BOOK COMPLIANCE / 18-MO MILESTONE / SHARED REMEDIATION COST" → "kraljic-strategic playbook applied with plambeck-scope-3 metrics. supplier accepts open-book on the regulated dimension, you fund the audit, both invest in remediation. multi-year alignment preserved; reg flank cleared."
        - **warn 0** "CONTINGENT TERMINATION CLAUSE / 6-MO TRIGGER" → "fisher-ury BATNA play but at the wrong tier. strategic suppliers respond to staged collaboration, not termination triggers. you've signaled transactional behavior to a strategic relationship."
        - **warn -1** "PRICE CONCESSION ONLY · NO STRUCTURAL CHANGE" → "you treated a structural problem with a transactional remedy. the next scandal cycle

(and the literature predicts there will be one) reopens this from a worse position."

- **D3** "INTERNAL COMMUNICATION.":
  - **good +2** "ONE-PAGE DECISION MEMO · NAMED FRAMEWORKS" → "CEO and board can defend the decision. the explicit framework citations (kraljic, tang, plambeck, sheffi) signal rigor and survive your eventual rotation out of the role."
  - **warn 0** "VERBAL BRIEFING · NARRATIVE FRAMING" → "works for one cycle. when the decision is challenged in board review or audit (and it will be), you defend it from memory rather than from a documented framework."
- **opt**: [0, 0, 0]
- **fr**: THE PRINCIPAL'S CRAFT IS NOT KNOWING ONE FRAMEWORK DEEPLY — IT IS KNOWING WHEN EACH OF THE 6–8 RELEVANT FRAMEWORKS APPLIES, IN WHAT ORDER THEY GATE EACH OTHER, AND WHERE THEY CONFLICT. THIS CASE PULLS FROM KRALJIC (PLACEMENT), TANG (CONCENTRATION ECONOMICS), FISHER-URY (BATNA), PLAMBECK (SCOPE 3 OPERATIONAL EXCELLENCE), AND SHEFFI (RESILIENCE) SIMULTANEOUSLY. THE FRIDAY DEADLINE FORCES INTEGRATION; THE REGULATORY EXPOSURE FORCES ORDERING. THE GRADUATION OUTPUT OF THE SCFC PROGRAM IS THE FLUENCY TO RUN THIS DECISION IN UNDER 60 MINUTES, WITH DEFENSIBLE CITATIONS, AND TO BE RIGHT MORE OFTEN THAN NOT.

Day 79 — Stochastic Inventory Capstone *(capstone)*

- **dom**: 3 (INVENTORY)
- **f**: (Q,r) Policy with Random Demand & Lead Time
- **o**: Hadley & Whitin (1963); Federgruen & Zheng (1992)
- **cite**: Hadley, G. & Whitin, T. M. *Analysis of Inventory Systems*. Prentice-Hall, 1963.
- **sum**: Real reorder policies face both demand variability and lead-time variability. The (Q, r) model — order Q when inventory hits r — has a closed-form optimum that the principal should derive once.
- **dur**: 18
- **cap**: true
- **deriv**: full
- **content** (lecture+derivation):

- **derivation steps**:

  1. **Setup**: Demand per period D ~ Normal(μ, σ_D²); lead time L ~ Normal($\bar{L}$, σ_L²); reorder point r; order quantity Q. Service level target 1 – α.

  2. **Demand during lead time** (DDLT):

     ```
     E[DDLT] = μ · L̄
     Var[DDLT] = L̄ · σ_D² + μ² · σ_L²     ← key identity
     SD[DDLT] = √(L̄ · σ_D² + μ² · σ_L²)
     ```

  3. **Reorder point with safety stock**:

     ```
     r = E[DDLT] + z_α · SD[DDLT]
       = μ · L̄ + z_α · √(L̄ · σ_D² + μ² · σ_L²)
     ```

     where z_α is the inverse standard normal at service level 1 – α.

  4. **Order quantity** (separately, via EOQ):

     ```
     Q* ≈ √(2DS / H)          (Wilson)
     ```

     Q and r are *separately* optimized in the standard formulation; joint optimization (Federgruen-Zheng) is more accurate but costlier to compute.

  5. **Numerical example**:

     - μ = 200/wk, σ_D = 40, $\bar{L}$ = 4 wk, σ_L = 1 wk, target service 95% (z = 1.645)
     - E[DDLT] = 800
     - Var[DDLT] = 4 · 40² + 200² · 1² = 6,400 + 40,000 = 46,400
     - SD[DDLT] = 215
     - r = 800 + 1.645 · 215 = 800 + 354 = **1,154 units**

  6. **The lead-time-variability dominance**: notice that 40,000 (lead time variance contribution) > 6,400 (demand variance contribution). When lead time is variable, *it is the dominant source of uncertainty*. Reducing lead time variance is typically cheaper than reducing demand variance, and matters more.

- **fr**: THE (Q, R) MODEL IS THE WORKHORSE OF MULTI-PERIOD INVENTORY MANAGEMENT. THE NON-OBVIOUS LESSON FROM THE DERIVATION IS THAT *LEAD-TIME VARIABILITY OFTEN DOMINATES DEMAND VARIABILITY* IN DRIVING SAFETY STOCK. THIS IS WHY SUPPLIER PERFORMANCE METRICS (ON-TIME

DELIVERY, LEAD-TIME CONSISTENCY) MATTER SO MUCH TO INVENTORY ECONOMICS. THE PRINCIPAL'S MOVE WHEN STOCK LEVELS ARE TOO HIGH IS USUALLY NOT TO IMPROVE FORECASTING — IT IS TO TIGHTEN LEAD-TIME VARIANCE WITH SUPPLIERS. THE MATH SAYS THE LEVERAGE IS THERE.

Day 80 — Capstone · The Principal's Synthesis *(capstone)*

- **dom**: 12 (APPLIED)
- **f**: Cross-Domain Integration · The Graduation Frame
- **o**: Synthesis of Stanford OIT/MS&E canon
- **cite**: Comprehensive curriculum integration.
- **sum**: The graduation day. A reflection on how the 80 days fit together, what the canon collectively teaches, and what the principal's craft looks like in practice. Not a case — a directed reflection with prompts and a final framework map.
- **dur**: 15
- **cap**: true
- **content**:
  - **lecture-style** with checkpoints. The user is asked to recall, in succession:
    1. Three frameworks they applied this past month.
    2. The framework they most often forget to apply.
    3. A decision they made before this program that, with hindsight, used the wrong framework.
    4. A future decision (named) where they'll consciously apply at least three frameworks.
  - The output is a "principal's map" — a single screen showing the 18 domains and the user's ranked self-assessment of fluency in each.
  - **fr**: A STANFORD MS IN OPERATIONS DOES NOT MAKE YOU A PRINCIPAL. WHAT IT DOES IS GIVE YOU FLUENCY IN THE CANON SO THAT WHEN A COMPLEX DECISION ARRIVES, YOU REACH FOR THE RIGHT FRAMEWORK INSTEAD OF YOUR INSTINCTS. SCFC HAS DELIVERED THE SAME FLUENCY IN 80 DAILY SESSIONS. THE WORK FROM HERE — AND IT IS THE WORK THAT ACTUALLY MAKES A PRINCIPAL — IS APPLYING THE FRAMEWORKS UNDER REAL CONSTRAINTS, WITH REAL CONSEQUENCES, OVER MANY YEARS. THIS IS WHERE THE DEGREE ENDS

AND THE CRAFT BEGINS.

---

4 · Glossary expansion

Add 30 terms to the existing 50, for a total of 80. Render as before in 2-column grid.

```
const NEW_GLOSSARY_TERMS = [
  // FOUNDATIONS / STRATEGY (extending)
  ['TRIPLE-A',       'Hau Lee\'s framework: Agility, Adaptability, Alignment. Bes
  ['VALUE CAPTURE',  'Brandenburger-Stuart: the share of created value a firm app
  ['CAPABILITY',     'Pisano: a firm-specific bundle of routines and assets that

  // OR / NETWORKS (new domain)
  ['p-MEDIAN',       'MILP formulation: choose p facilities from candidates to mi
  ['p-CENTER',       'MILP variant: minimize the maximum distance any customer tr
  ['SET COVERING',   'Minimum number of facilities such that every demand point i
  ['MILP',           'Mixed-Integer Linear Programming. Workhorse of network desi
  ['DUAL VARIABLE',  'In LP, the shadow price of a constraint. Reveals which cons
  ['NP-HARD',        'Computational class. VRP, p-median, scheduling — all NP-har
  ['SAVINGS ALGO',   'Clarke-Wright heuristic for VRP. Combines routes by selecti
  ['LITTLE\'S LAW',  'L = λ × W. Average inventory = throughput × time-in-system.
  ['M/M/1',          'Single-server queue with Poisson arrivals and exponential s
  ['ERLANG C',       'Multi-server queueing formula for call center / dock staffi

  // SUSTAINABILITY (new domain)
  ['SCOPE 1/2/3',    'GHG emissions accounting: 1=direct (boilers), 2=purchased e
  ['CBAM',           'EU Carbon Border Adjustment Mechanism (2023). Prices import
  ['EU ETS',         'EU Emissions Trading System. ~€80–100/tonne CO₂ as of 2026.
  ['SBTi',           'Science Based Targets initiative. Validates corporate emiss
  ['CIRCULARITY',    'Designing supply chains for reuse, repair, remanufacturing.

  // GEOPOLITICS (new domain)
  ['FRIEND-SHORING', 'Yellen 2022. Relocating production to politically aligned c
  ['DECOUPLING',     'Reducing supply chain interdependence with geopolitical riv
  ['CHIPS ACT',      '$200B+ US semiconductor industrial policy (2022). Largest p
  ['IRA',            'Inflation Reduction Act (2022). $370B for clean energy + EV
  ['SECTION 301',    'US trade law authorizing tariffs on China. ~25% on most cat

  // HEALTHCARE / PHARMA (new domain)
  ['DSCSA',          'Drug Supply Chain Security Act. Mandates unit-level seriali
  ['EPCIS',          'Electronic Product Code Information Services. The data exch
  ['COLD CHAIN',     'Temperature-controlled supply chain (typically 2–8°C for va
  ['FDA 483',        'FDA notice of objectionable conditions during inspection. O
```

```
  // SEMICOND (new domain)
  ['EUV',            'Extreme Ultraviolet Lithography. ASML monopoly. Gates ≤7nm
  ['FOUNDRY',        'Pure-play semiconductor manufacturer (TSMC, Samsung Foundry
  ['ALLOCATION',     'When fab capacity is constrained, the share each customer r

  // BEHAVIOR (new domain)
  ['BEER GAME',      'Sterman 1989 simulation showing bullwhip emerges even with
  ['ANCHORING',      'Tversky-Kahneman 1974. First number framing dominates subse
];
```

Total glossary after merge: 80 terms.

---

5 · Reading list expansion

Add 18 entries. Schema unchanged from v2. The CANON tier is reserved for the most foundational works; CORE is strong supporting research. v3 adds a third tier — **EMERGING** — for post-2018 work where the canon hasn't fully consolidated. EMERGING renders with a blue tier badge (color `#7AB8E0`).

```
const NEW_READING = [
  // Triple-A & Stanford operations canon
  ['CANON',    'Lee, H. L.',                '"The Triple-A Supply Chain."',
  ['CORE',     'Lee, H. L. & Tang, C. S.',  '"Socially and Environmentally Respo
  ['CANON',    'Brandenburger, A. M. & Stuart, H. W.', '"Value-based Business Str
  ['CORE',     'Pisano, G. P.',             '"Toward a Prescriptive Theory of Dy

  // OR / Networks
  ['CANON',    'Daskin, M. S.',             'Network and Discrete Location: Mode
  ['CORE',     'Toth, P. & Vigo, D.',       'Vehicle Routing: Problems, Methods,
  ['CANON',    'Little, J. D. C.',          '"A Proof for the Queuing Formula: L
  ['CORE',     'Hopp, W. J. & Spearman, M. L.', 'Factory Physics',

  // Sustainability
  ['CANON',    'Plambeck, E. L.',           '"Reducing greenhouse gas emissions
  ['CORE',     'Lee, H. L., Plambeck, E. L. & Yatsko, P.', '"Embracing Green in C
  ['CORE',     'WRI / WBCSD',               'Corporate Value Chain (Scope 3) Acc
  ['EMERGING', 'European Commission',       'Carbon Border Adjustment Mechanism,

  // Geopolitics
  ['EMERGING', 'Witt, M. A. et al.',        '"Decoupling in international busine
  ['EMERGING', 'Allen, G. C. (CSIS)',       'Sourcing Requirements and U.S. Tech
  ['CORE',     'Mazzucato, M.',             'The Entrepreneurial State',

  // Healthcare
```

```
  ['CORE',     'Scheinker, D. & Brandeau, M. L.', '"Healthcare operations managem
  ['EMERGING', 'FDA',                             'Drug Supply Chain Security Act: Imp

  // Behavioral Operations
  ['CANON',    'Sterman, J. D.',                  '"Modeling Managerial Behavior: Misp
  ['CANON',    'Croson, R. & Donohue, K.',        '"Behavioral Causes of the Bullwhip
  ['CORE',     'Gino, F. & Pisano, G.',           '"Toward a Theory of Behavioral Oper
  ['CORE',     'Galinsky, A. D. & Mussweiler, T.', '"First Offers as Anchors."',
];
```

Total reading list after merge: 40+ entries. Re-sort by day in the Reading view.

5.1 Tier badge styling

Add EMERGING tier:

```
.read-row .tier.emerging {
  color: #7AB8E0;
  border-color: rgba(122,184,224,0.5);
}
```

---

6 · Schema additions

6.1 Day type

Extend the existing schema. Existing days continue to work without modification (new fields are optional).

```
type Day = {
  d: number;
  w: number;              // week number (1..16 for v3 — see §6.4)
  dom: number;            // 1..18
  t: string;
  f: string;
  o: string;
  cite: string;
  sum: string;
  scn: Scenario | null;
  // NEW IN V3
  dur?: number;           // estimated duration in minutes (default 5; capstones 1
  cap?: boolean;          // capstone flag — renders gold border, "CAPSTONE" badge
  deriv?: Derivation;     // optional full derivation panel (capstone-only typical
};
```

```
type Derivation = {
  // Renders in a new panel below the quant panel
  steps: Array<{
    n: string;           // step number (e.g., '1', '2.a')
    title: string;       // step title (e.g., 'First-order condition')
    body: string;        // markdown-allowed body (line breaks preserved)
    math?: string;       // optional formula block (rendered as IBM Plex Serif it
  }>;
};
```

6.2 Reading entry

```
type ReadingEntry = [
  tier: 'CANON' | 'CORE' | 'EMERGING',  // EMERGING is new
  author: string,
  work: string,
  publisher: string,
  topic: string,
  day: number,
  // NEW IN V3 (optional, additive)
  tag?: string  // e.g. 'derivation', 'case-study', 'survey' — surfaces in palett
];
```

6.3 Derivation panel rendering

CSS for the new derivation panel:

```
.deriv {
  margin-top: 14px; padding: 14px;
  border: 1px solid rgba(255,192,128,0.55);
  background: rgba(255,192,128,0.04);
  position: relative; z-index: 3;
}
.deriv .dh {
  font-family: 'Major Mono Display'; font-size: 9px;
  letter-spacing: 0.24em; color: #FFC080;
  margin-bottom: 8px;
  display: flex; align-items: center; gap: 8px;
}
.deriv .dh::before {
  content: '∫'; font-family: 'IBM Plex Serif';
  font-size: 14px; font-style: italic;
}
.deriv .step {
```

```
  margin-bottom: 12px; padding-bottom: 10px;
  border-bottom: 1px dashed rgba(255,192,128,0.2);
}
.deriv .step:last-child { border-bottom: none; }
.deriv .step .sn {
  font-family: 'Major Mono Display'; font-size: 10px;
  color: #FFC080; letter-spacing: 0.18em;
  margin-bottom: 4px;
}
.deriv .step .stitle {
  font-family: 'IBM Plex Serif'; font-size: 14px;
  color: #C8F0A0; font-weight: 500;
  margin-bottom: 6px;
}
.deriv .step .sbody {
  font-family: 'IBM Plex Mono'; font-size: 12px;
  line-height: 1.55; color: rgba(184,224,144,0.9);
  margin-bottom: 6px;
}
.deriv .step .smath {
  font-family: 'IBM Plex Mono'; font-size: 12.5px;
  color: #C8F0A0; padding: 8px 10px;
  background: rgba(0,0,0,0.25);
  border-left: 2px solid #FFC080;
  white-space: pre-wrap; line-height: 1.5;
  margin: 6px 0;
}
```

Render derivation panels *after* the quant panel (if both exist). Typewriter effect *does not apply* to derivations — they render statically once the brief is complete. This is a deliberate design choice: derivations are reference material, not theater.

6.4 Week numbers

v3 adds 4 weeks (Foundation 60 keeps weeks 1–12; Mastery +20 spans weeks 13–16).

```
WEEK 13 — NETWORKS + ADVANCED INVENTORY
  Day 61 — Triple-A Supply Chain (case)
  Day 62 — Newsvendor Derivation (capstone)
  Day 63 — Network Optimization Intuition (case)
  Day 64 — Vehicle Routing Problem (lecture)
  Day 65 — Queueing Intuition · Little's Law (lecture)

WEEK 14 — SUSTAINABILITY + GEOPOLITICS
  Day 66 — Queueing Capstone (capstone)
  Day 67 — Sustainability & Operations (case)
```

```
  Day 68 — Carbon Pricing & Cap-and-Trade (lecture)
  Day 69 — Sustainability Capstone · Carbon Math (capstone)
  Day 70 — Geopolitical Supply Chain (case)

WEEK 15 — HEALTHCARE + SEMICONDUCTORS
  Day 71 — CHIPS Act & Industrial Policy (lecture)
  Day 72 — Healthcare Supply Chain (case)
  Day 73 — DSCSA & Pharma Compliance (lecture)
  Day 74 — Semiconductor Supply Chain (case)
  Day 75 — Lithography Monopoly · ASML (lecture)

WEEK 16 — BEHAVIORAL + CAPSTONE INTEGRATION
  Day 76 — Behavioral Operations (case)
  Day 77 — Anchoring & Negotiation (lecture)
  Day 78 — Capstone · Multi-Domain Synthesis (capstone)
  Day 79 — Stochastic Inventory Capstone (capstone)
  Day 80 — Capstone · The Principal's Synthesis (capstone)
```

---

7 · UI additions

7.1 Capstone day badge

A capstone day card in the curriculum grid gets a gold accent border and a small `CAPSTONE` ribbon in the top-right.

```
.curr-day.capstone {
  border-color: rgba(255,192,128,0.7);
  background: rgba(255,192,128,0.06);
  position: relative;
}
.curr-day.capstone::after {
  content: 'CAPSTONE';
  position: absolute; top: -1px; right: -1px;
  background: #FFC080; color: #1A1814;
  font-family: 'IBM Plex Mono'; font-size: 7px;
  letter-spacing: 0.2em; padding: 2px 5px;
  border-radius: 0 0 0 3px;
}
.curr-day.capstone .dnum::after { content: ' ⊕'; color: #FFC080; }
```

7.2 Duration display

On the home view for a capstone day, show duration prominently:

```
DAY 66 / 80 · CAPSTONE · ~18 min
```

Standard days continue to show only the day number (5-min sessions are the default and don't need a duration call-out).

7.3 Tier toggle in nameplate

Add a small toggle in the nameplate area indicating which program tier the user is in:

```
[FOUND·60]  [MAST·+20]
```

- Foundation 60 is always available.
- Mastery +20 is locked until day 60 is complete; lock state shows a small 🔒 (no emoji — use Unicode character ▪) and a tooltip "Complete Foundation 60 to unlock."

Toggling between tiers does not change `state.currentDay`; it filters the curriculum view. The "Today" view always loads `state.currentDay`, regardless of tier toggle.

7.4 Mastery completion celebration

When the user completes day 80, render a brief LCD sequence:

```
// program complete.
//
// ──────────────────────────────────
//
// 80 days · 18 domains · 30 frameworks
// 14 cases · 8 capstones · 80 glossary terms
// 40+ canonical citations
//
// you have completed the foundation
// of a supply chain masters fluency.
//
// the work from here is application.
//
// ──────────────────────────────────
```

Then show the principal's map (§3, Day 80 deliverable) and the user's domain strength bars at maximum visualization.

---

8 · Migration from v2

On first load with v3 code, look for the existing v2 key and migrate.

```
function migrateFromV2() {
  const v2raw = localStorage.getItem('scfc-masters-v2');
  if (!v2raw) return null;
  try {
    const v2 = JSON.parse(v2raw);
    return {
      version: 3,
      streak: v2.streak || 0,
      currentDay: v2.currentDay || 1,
      lastCompletedDate: v2.lastCompletedDate || null,
      completions: v2.completions || [],
      domains: {
        ...v2.domains,
        // Initialize new domains (13–18) at strength 0
        13: { strength: 0 },
        14: { strength: 0 },
        15: { strength: 0 },
        16: { strength: 0 },
        17: { strength: 0 },
        18: { strength: 0 }
      },
      tier: (v2.currentDay || 1) > 60 ? 'mastery' : 'foundation'
    };
  } catch (e) { return null; }
}

function loadState() {
  // v3 first
  const v3raw = localStorage.getItem('scfc-masters-v3');
  if (v3raw) {
    try { return Object.assign(defaultState(), JSON.parse(v3raw)); }
    catch (e) { /* fall through */ }
  }
  // try v2 migration
  const migrated = migrateFromV2();
  if (migrated) {
    localStorage.setItem('scfc-masters-v3', JSON.stringify(migrated));
    // optionally remove v2 key after successful migration:
    // localStorage.removeItem('scfc-masters-v2');
    return migrated;
  }
  return defaultState();
}
```

```
function defaultState() {
  return {
    version: 3,
    streak: 0,
    currentDay: 1,
    lastCompletedDate: null,
    completions: [],
    domains: Object.fromEntries(DOMAINS.map(d => [d.n, { strength: 0 }])),
    tier: 'foundation'
  };
}
```

The v2 key is *not deleted* after migration unless the user opts in via a palette command. This is conservative — if migration fails downstream, the v2 data is recoverable.

---

9 · Acceptance checklist (v3-specific additions)

Add to v2 checklist:

9.1 Domain expansion

- VU meter renders 18 rows.
- Mobile: VU meters scroll vertically without breaking layout.
- Domain names 13–18 render correctly in the curriculum grid week labels.

9.2 Curriculum

- All 80 days reachable from the curriculum grid (with tier toggle to Mastery).
- Day 60 completion unlocks Mastery tier toggle.
- Capstone days render with gold ribbon and accent border.
- Day 60 vs Day 80 progress display correctly per tier.

9.3 Capstone days

- All 8 capstone days (62, 66, 69, 78, 79, 80, plus the v2 cases reclassified if any) display the duration call-out.
- Derivation panels render for days that have them (62, 66, 69, 79).
- Derivations are static (no typewriter).

- Math blocks use monospace and the gold left-border style.

9.4 Glossary

- All 80 terms render in the glossary view.
- New terms are searchable in the palette.

9.5 Reading list

- All 40+ entries render.
- EMERGING tier renders in blue (#7AB8E0).
- Reading list re-sorts by day with new entries integrated.

9.6 Migration

- User with v2 state loads into v3 with progress preserved.
- User with no prior state loads day 1 in foundation tier.
- Streak, completions, domain strengths from v2 are intact post-migration.
- New domain strengths (13–18) initialize at 0.

9.7 Day 80 completion

- Day 80 completion triggers the program-complete LCD sequence.
- Principal's map renders.
- No subsequent days unlock (program is complete; replays are accessible from curriculum).

---

10 · Out of scope (still — for v3)

The v2 list still applies. Additional v3 non-goals:

- **Live data integration** (e.g., real diesel prices, real CBAM credit prices). Scenarios use authored numbers; updating them is a v4 concern.
- **Adaptive difficulty.** All 80 days are linear and authored. No spaced retrieval, no skill-gap detection, no personalized ordering.
- **Multi-language.** English only.

- **Group mode / leaderboard.** Single user, no comparison features.
- **Certificate / badge generation.** The product respects the user's privacy and doesn't generate shareable artifacts.

---

11 · Definition of done — v3

The v3 build is complete when:

1. All v2 acceptance criteria still pass.
2. Every item in §9 above passes.
3. A user with existing v2 progress can launch v3 and find their state intact.
4. A new user can complete days 1–80 in sequence without encountering a broken state.
5. The reference Stanford alignment is defensible: every new domain has a citation linking it to either an MS&E or OIT course or a faculty body of work, and the canonical literature for each domain is referenced verbatim.
6. A senior procurement leader (Kyle profile) can finish all 80 days in 8–12 weeks with daily 5-minute sessions and 8 longer (15–20 min) capstone sessions, and emerge with defensible fluency in the supply chain canon at Stanford MS level.

That is the bar. Build to it.